PROJECT PORTFOLIO · REMOTE SENSING + MACHINE LEARNING

Iceberg Alley SAR

Finding icebergs off Newfoundland in Sentinel-1 radar with THOR, a SAR foundation model, benchmarked against a classical detector.

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Independent project, 2026
Code: github.com/natali-kovalski/iceberg-alley-sar
Live demo: natali-kovalski.github.io/iceberg-alley-sar



The live viewer, 8 May 2025: THOR detections (cyan rings), CA-CFAR baseline (orange dots), pack ice (blue), likely floes in grey.

84%
of Ice Patrol-labelled icebergs found by THOR on four held-out passes (431 of 512)

+10 pts
over a classical CA-CFAR (74%) with the same number of detections

11 of 14
large bergs found by THOR; the CFAR's ice-strip rule kept 4

769
trained parameters on top of the frozen foundation model

The problem

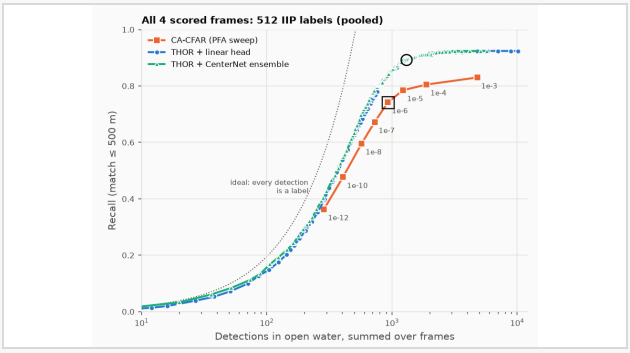
Each spring, icebergs from Greenland drift down the Labrador Current past northeast Newfoundland, into shipping lanes, fishing grounds and offshore platform areas. Aircraft patrols are costly and grounded by fog. Synthetic aperture radar (SAR) images the ocean through cloud and darkness. The question: can a foundation model, used off the shelf, find these bergs in free Sentinel-1 data better than the classical detector?

Approach

1. **Search and calibrate.** HH/HV scenes from ASF, terrain-corrected by HyP3 RTC at 10 m.
2. **Mask.** Land from OpenStreetMap plus 500 m; pack ice from cross-pol (HV) texture on 200 m cells.
3. **Detect with THOR.** The frozen encoder turns each scene into features; a linear head trained on Ice Patrol labels (2019–2020) turns them into an iceberg heatmap.
4. **Baseline.** A cell-averaging CFAR on HV with size, ice-strip and HH checks, on the same masks.
5. **Publish.** One GeoJSON schema for both, then CZML for a CesiumJS viewer on GitHub Pages.

A fair head-to-head

Both detectors ran on four April 2019 passes that were never used for training or tuning, with the same masks, the same Ice Patrol labels and the same 500 m matching. Precision needs ship positions (AIS), so I compared **recall at the same number of detections**. THOR leads on every pass and reaches the CFAR's recall with 28% fewer detections. Small bergs are a tie, and THOR needs the classical pack-ice mask: on its own it fires on pack ice.



Recall vs. detections, 4 held-out passes, 512 bergs: THOR (blue, green) vs. the CA-CFAR over its false-alarm rates (orange).

Engineering practice

- **Modular Python package** with type hints, a CLI and 153 tests
- **No leakage:** held out by month, heads chosen on year folds
- **One config file**, lengths in metres across resolutions
- **Live viewer** with both layers, deployed by CI

Validation

Of four ground-truth sources (ice charts, aircraft sightings, Sentinel-2, same-pass Ice Patrol labels), only the last supports a per-berg score. The code reports the others as unresolved.

Limits and next steps

One month of one year; precision unknown without AIS. Next: AIS matching, THOR at 4×4 patches for small bergs, and THOR inside the one-command pipeline.

TOOLS
PyTorch · TerraTorch · Python · NumPy · SciPy · rasterio · GeoPandas · HyP3 · Vue 3 · CesiumJS

Data: modified Copernicus Sentinel data (2019–2026) · International Ice Patrol (NSIDC G00807) · North American Ice Service · THOR: FM4CS (Apache 2.0), Forgaard et al., IGARSS 2026 · © OpenStreetMap contributors